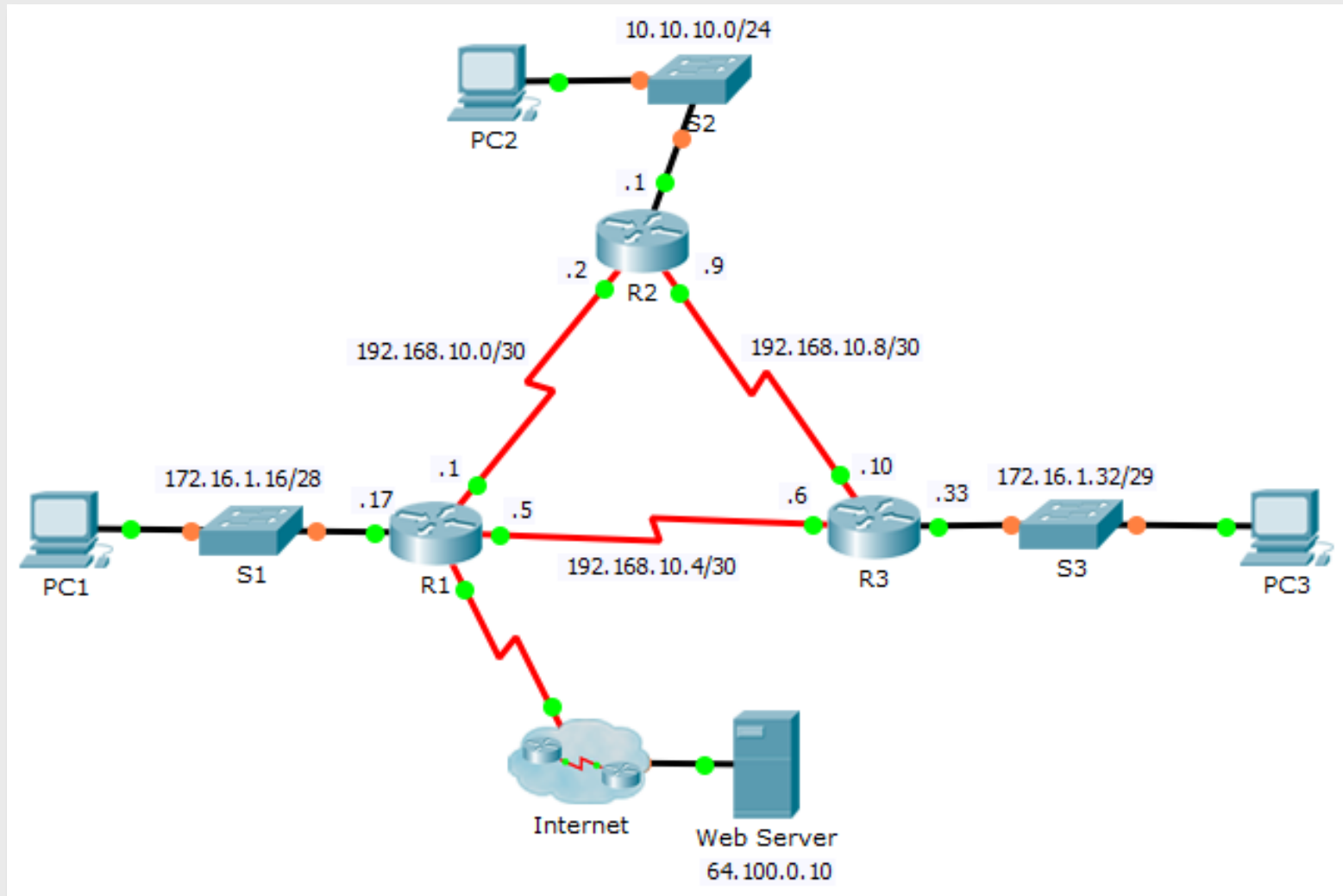


Lab 7

OSPF configuration

IOS OSPF configuration – single area

■ Reference topology

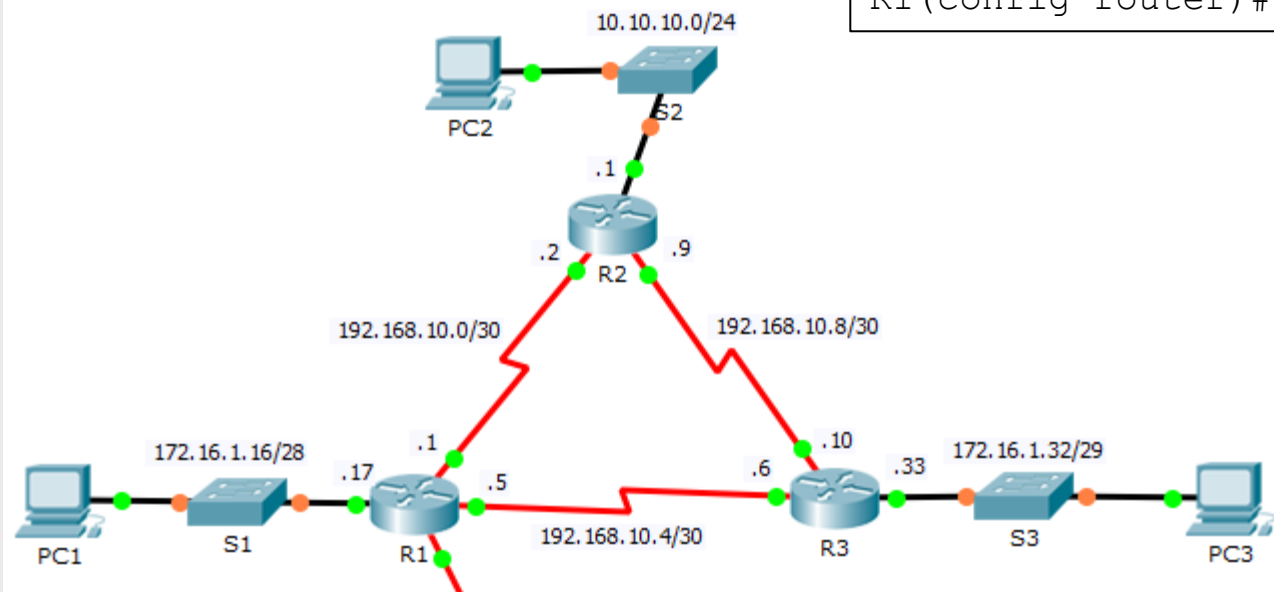


IOS OSPF configuration

```
Router(config)#router ospf process-id
```

- Enables OSPF routing
 - Default bandwidth: 1544 Kbps on all serial links

```
R1(config)#router ospf 1  
R1(config-router)#
```



IOS OSPF configuration

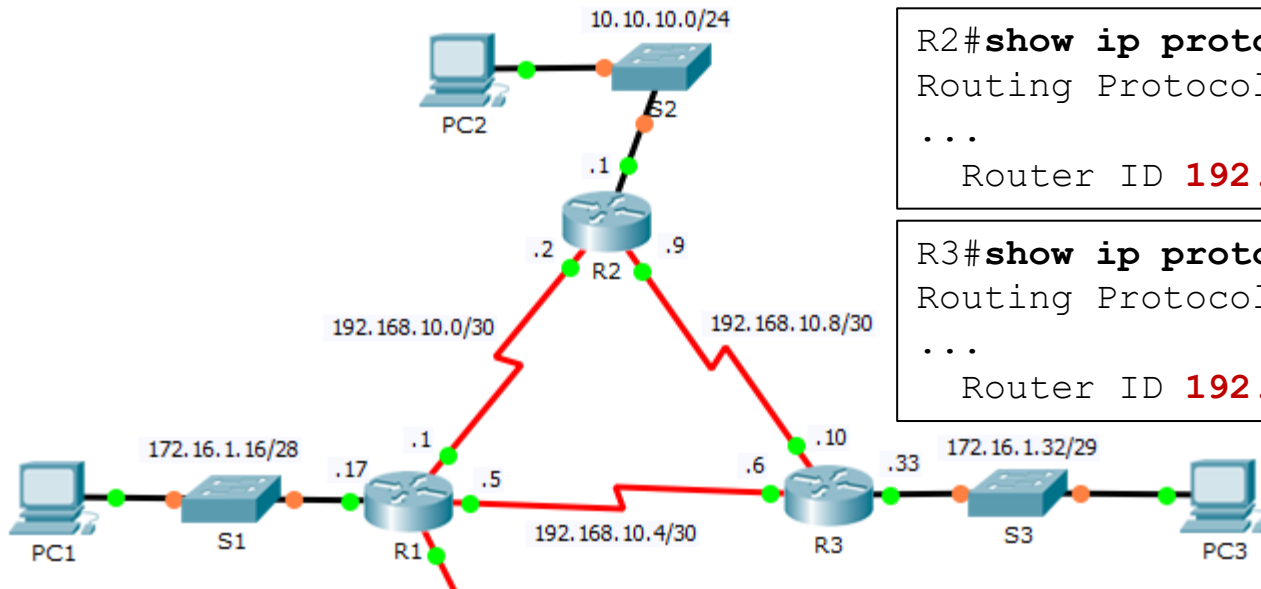
■ Determining the Router ID

1. IP address configured with the **router-id** command
2. The highest IP address of any of its loopback interfaces
3. The highest active IP address of any of its physical interfaces

```
R1#show ip protocols
Routing Protocol is "ospf 1"
...
Router ID 209.165.200.225
```

```
R2#show ip protocols
Routing Protocol is "ospf 1"
...
Router ID 192.168.10.9
```

```
R3#show ip protocols
Routing Protocol is "ospf 1"
...
Router ID 192.168.10.10
```



IOS OSPF configuration

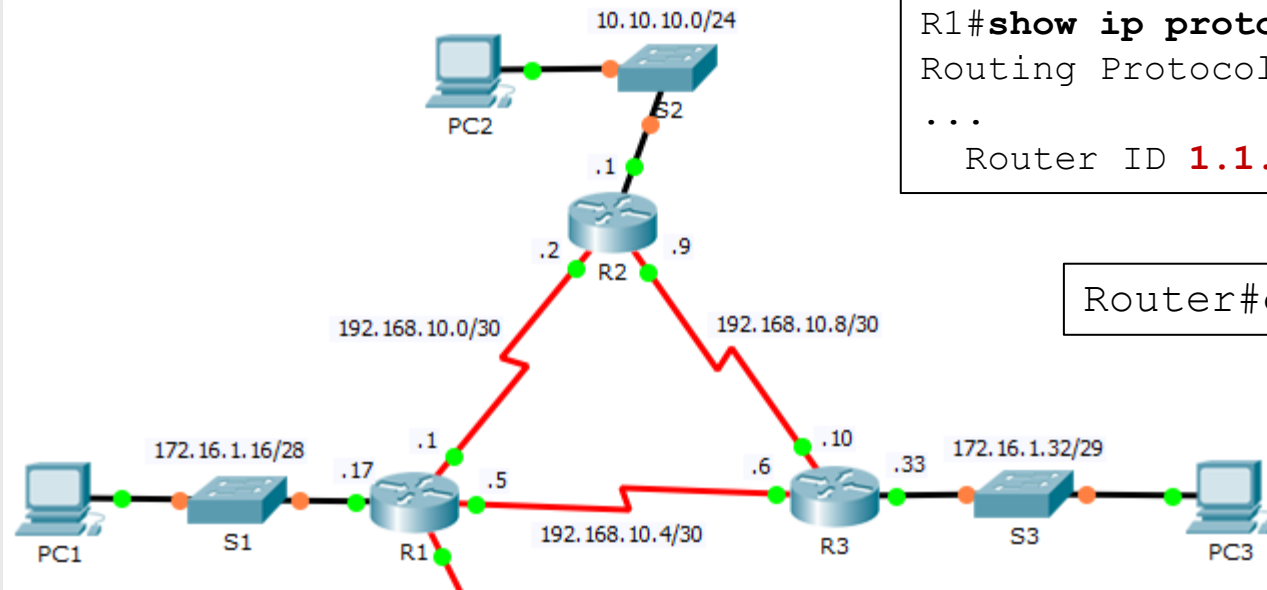
```
Router(config-router)#router-id ip-address
```

■ Determining the Router ID

1. IP address configured with the **router-id** command
2. The highest IP address of any of its loopback interfaces
3. The highest active IP address of any of its physical interfaces

```
R1(config)#router ospf 1  
R1(config-router)#router-id 1.1.1.1  
R1(config-router)#^Z  
R1#show ip protocols  
Routing Protocol is "ospf 1"  
...  
Router ID 1.1.1.1
```

```
Router#clear ip ospf process
```



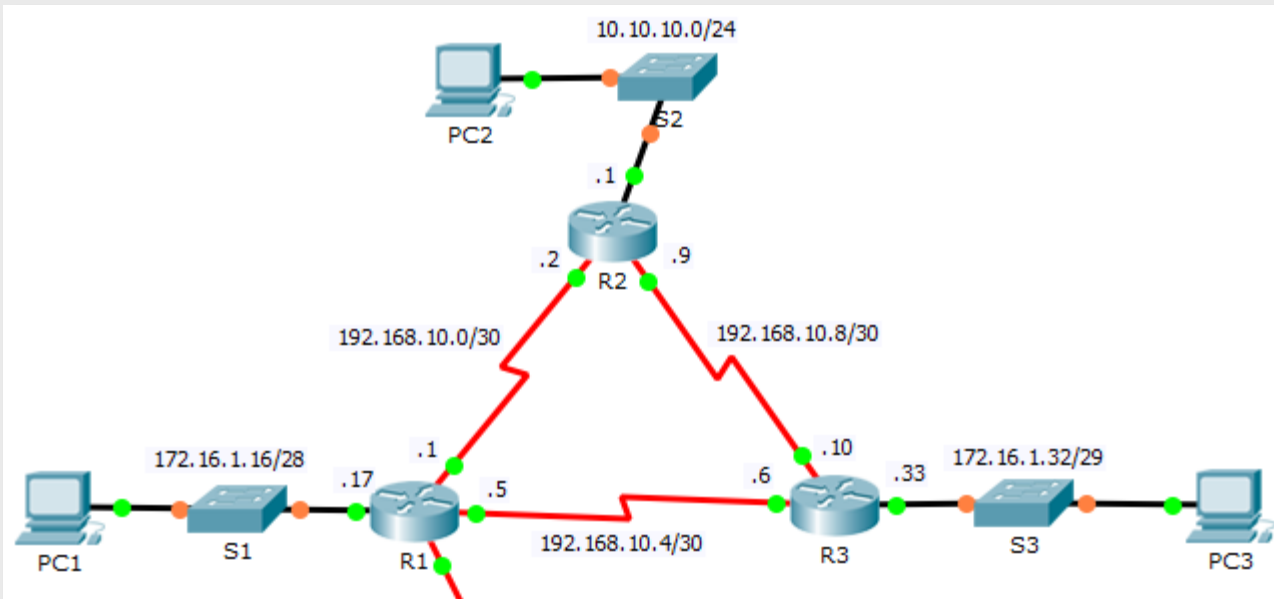
IOS OSPF configuration

```
Router(config-router)#router-id ip-address
```

■ Determining the Router ID

1. IP address configured with the **router-id** command
2. The highest IP address of any of its loopback interfaces
3. The highest active IP address of any of its physical interfaces

```
R1(config)#interface loopback 0  
R1(config-if)#ip address 1.1.1.1 255.255.255.255  
R1(config-if)#end
```

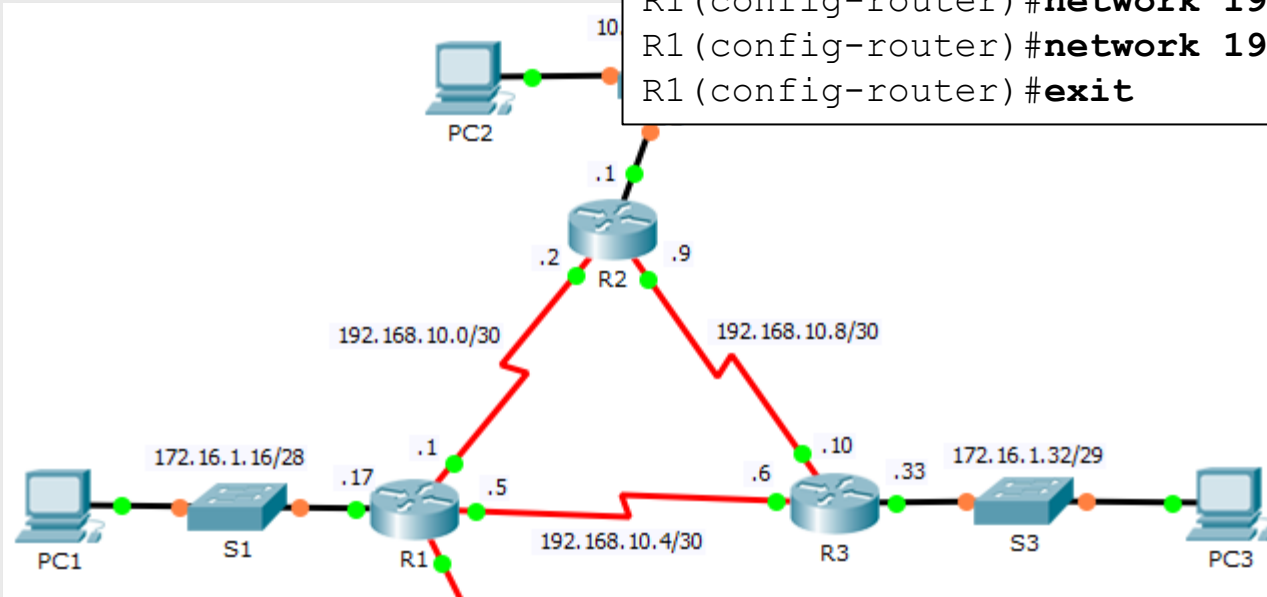


IOS OSPF configuration

```
Router(config-router)#network network-address wildcard-mask area area-id
```

- Determines which interfaces, per area, participate in the OSPF routing process
 - Any interface matching the network address will be enabled to send and receive OSPF packets
 - This network will be included in OSPF routing updates

```
R1(config)#router ospf 1  
R1(config-router)#network 172.16.1.16 0.0.0.15 area 0  
R1(config-router)#network 192.168.10.0 0.0.0.3 area 0  
R1(config-router)#network 192.168.10.4 0.0.0.3 area 0  
R1(config-router)#exit
```



IOS OSPF configuration

```
Router(config-router)#network intf-ip-address 0.0.0.0 area area-id
```

- Determines which interfaces, per area, participate in the OSPF routing process
 - The interface with the specified address will be enabled to send and receive OSPF packets
 - The network on the interface will be included in OSPF routing updates

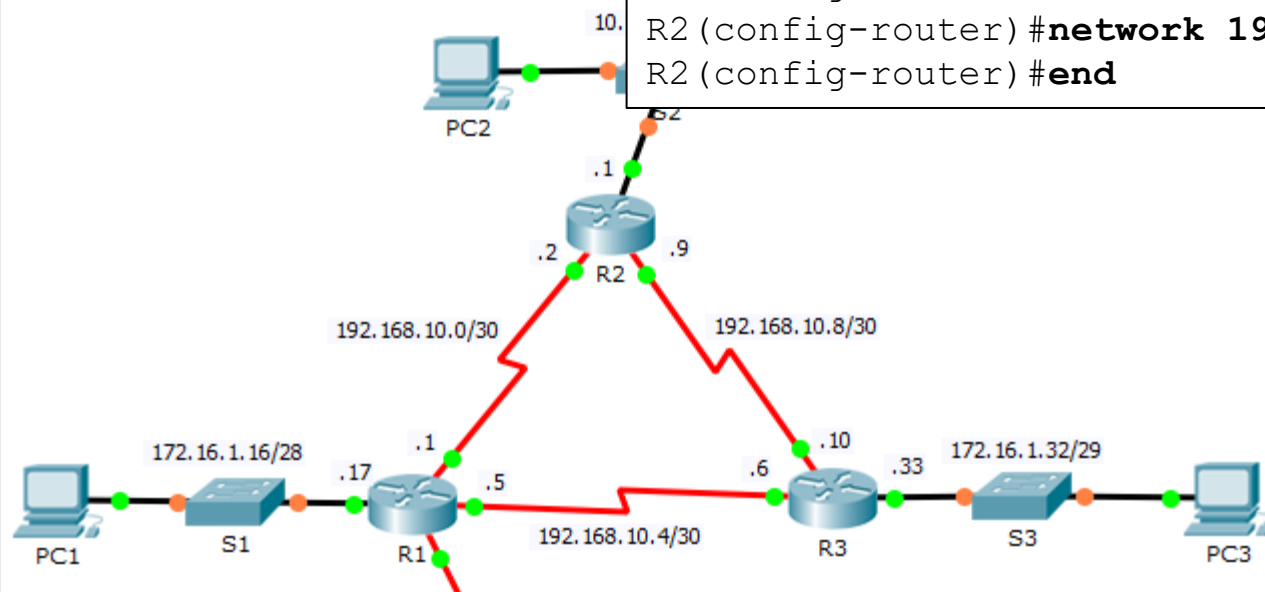
```
R2(config)#router ospf 1
```

```
R2(config-router)#network 10.10.10.1 0.0.0.0 area 0
```

```
R2(config-router)#network 192.168.10.2 0.0.0.0 area 0
```

```
R2(config-router)#network 192.168.10.9 0.0.0.0 area 0
```

```
R2(config-router)#end
```

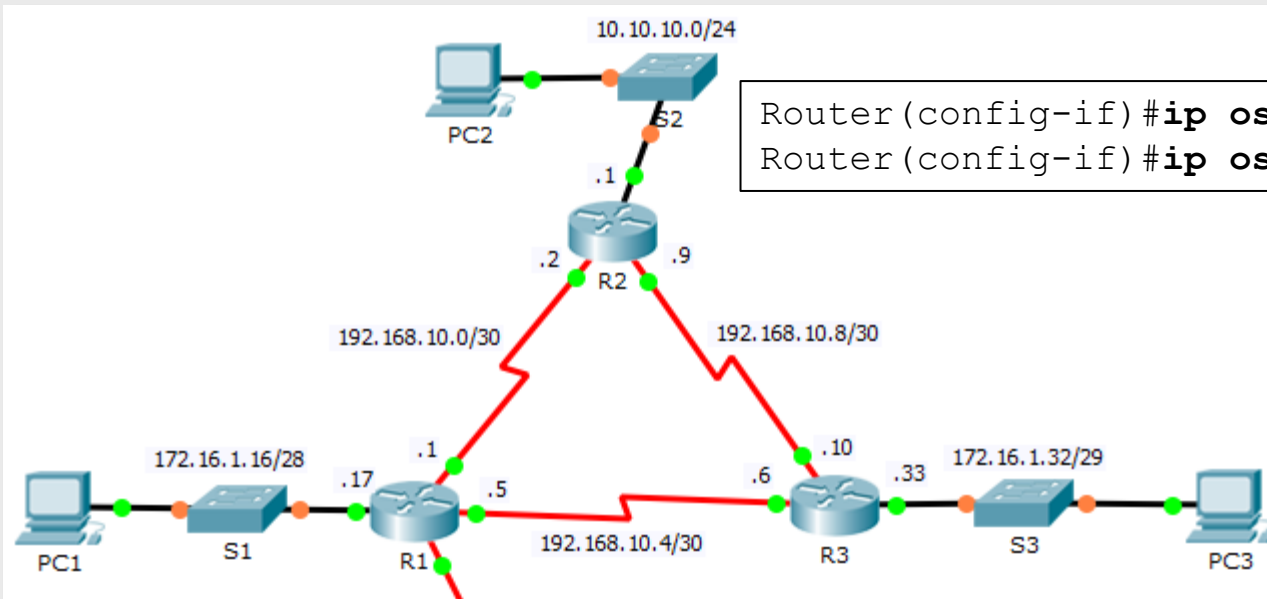


Verifying neighbor adjacency

- Two routers may not form an OSPF adjacency if
 - the subnet masks do not match
 - OSPF Hello or Dead Timers do not match
 - OSPF Network Types do not match
 - there is a missing or incorrect OSPF network command

```
R1#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
2.2.2.2	0	FULL/ -	00:00:33	192.168.10.2	Serial0/0/0
3.3.3.3	0	FULL/ -	00:00:32	192.168.10.6	Serial0/0/1



Verifying OSPF configuration

```
R1#show ip route
```

Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP

(**output omitted**)

10.0.0.0/24 is subnetted, 1 subnets

O 10.10.10.0 [110/65] via 192.168.10.2, 00:03:01, Serial0/0/0

172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks

C 172.16.1.16/28 is directly connected, FastEthernet0/0

O 172.16.1.32/29 [110/65] via 192.168.10.6, 00:03:01, Serial0/0/1

192.168.10.0/30 is subnetted, 3 subnets

C 192.168.10.0 is directly connected, Serial0/0/0

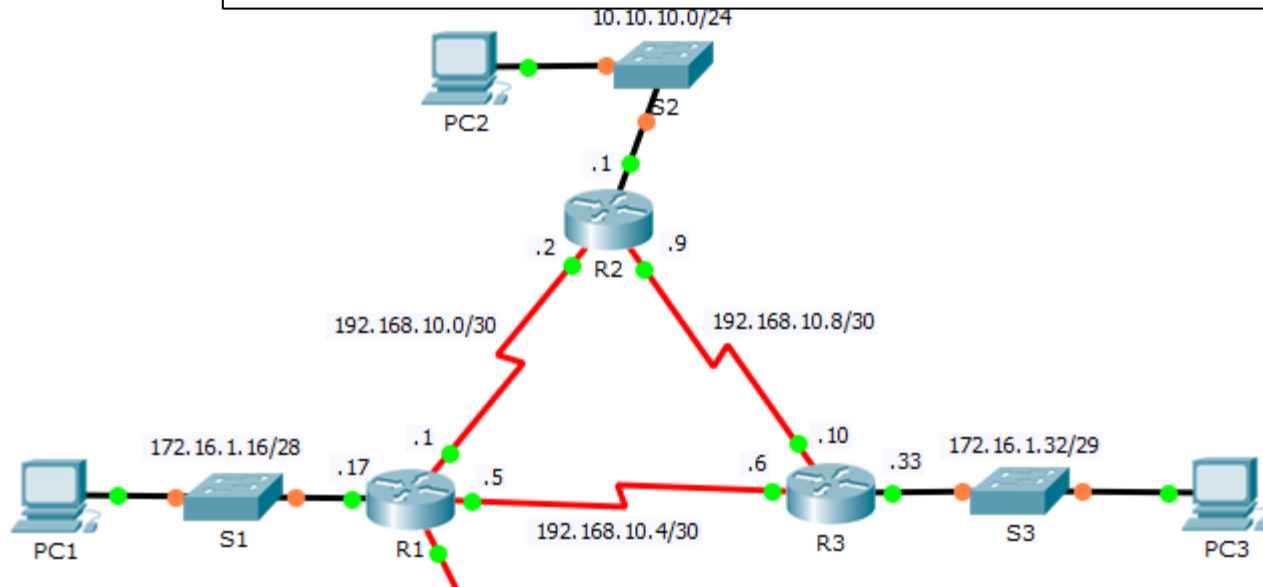
C 192.168.10.4 is directly connected, Serial0/0/1

O 192.168.10.8 [110/128] via 192.168.10.2, 00:03:01, Serial0/0/0

[110/128] via 192.168.10.6, 00:03:01, Serial0/0/1

209.165.200.0/30 is subnetted, 1 subnets

C 209.165.200.224 is directly connected, Serial0/1/0



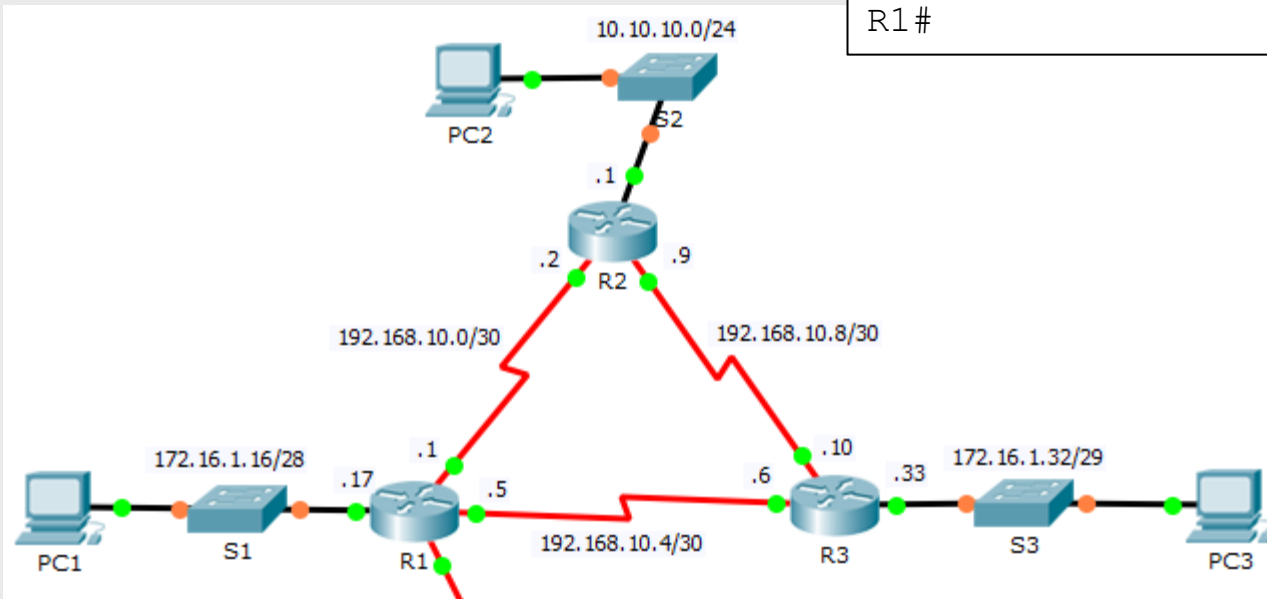
Passive interfaces

- By default, OSPF messages are forwarded out all OSPF-enabled interfaces

```
Router(config-router)#passive-interface interface-type interface-number
```

- Prevents the transmission of routing messages from the specified interface, but still allows that network to be advertised to other routers

```
R1(config)#router ospf 1  
R1(config-router)#passive-interface fa0/0  
R1(config-router)#^Z  
R1#
```



Verifying OSPF configuration

```
R1#show ip protocols
```

Routing Protocol is "ospf 1"

Outgoing update filter list for all interfaces is not set

Incoming update filter list for all interfaces is not set

Router ID 1.1.1.1

Number of areas in this router is **1**. 1 normal 0 stub 0 nssa

Maximum path: 4

Routing for Networks:

172.16.1.16 0.0.0.15 area 0

192.168.10.0 0.0.0.3 area 0

192.168.10.4 0.0.0.3 area 0

Passive Interface(s):

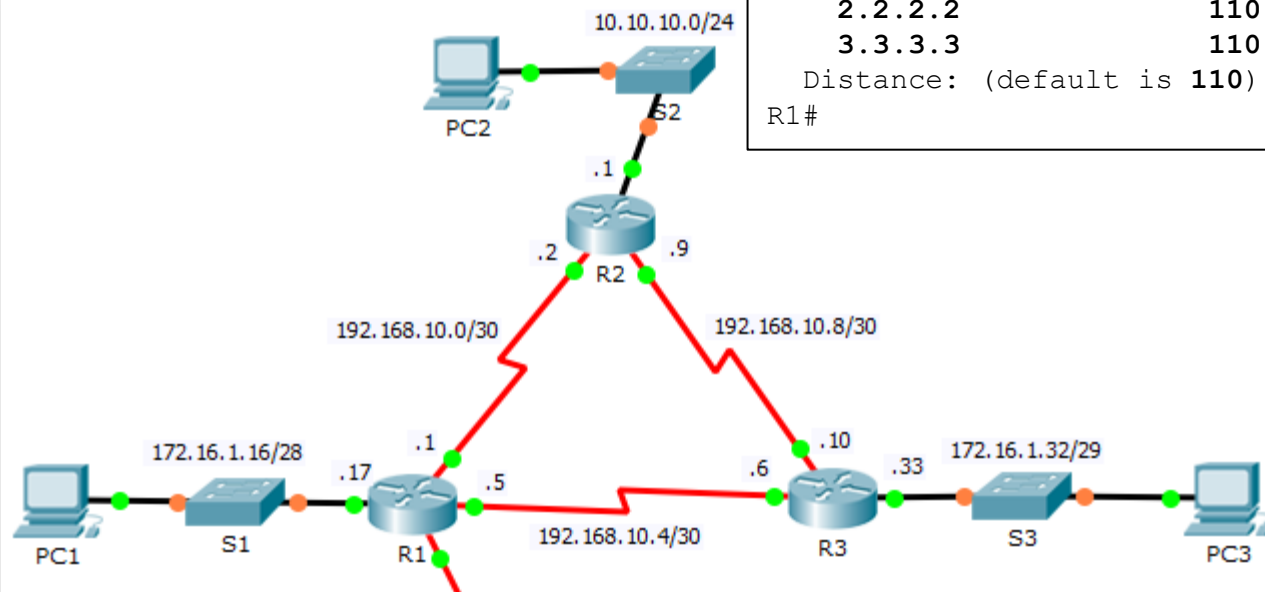
FastEthernet0/0

Routing Information Sources:

Gateway	Distance	Last Update
1.1.1.1	110	00:18:45
2.2.2.2	110	00:18:25
3.3.3.3	110	00:18:25

Distance: (default is 110)

```
R1#
```



Verifying OSPF configuration

```
R1#show ip ospf
```

```
Routing Process "ospf 1" with ID 1.1.1.1
```

```
Supports only single TOS(TOS0) routes
```

```
Supports opaque LSA
```

```
SPF schedule delay 5 secs, Hold time between two SPFs 10 secs
```

```
Minimum LSA interval 5 secs. Minimum LSA arrival 1 secs
```

```
Number of external LSA 0. Checksum Sum 0x000000
```

```
Number of opaque AS LSA 0. Checksum Sum 0x000000
```

```
Number of DCbitless external and opaque AS LSA 0
```

```
Number of DoNotAge external and opaque AS LSA 0
```

```
Number of areas in this router is 1. 1 normal 0 stub 0 nssa
```

```
External flood list length 0
```

```
Area BACKBONE(0)
```

```
Number of interfaces in this area is 3
```

```
Area has no authentication
```

```
SPF algorithm executed 9 times
```

```
Area ranges are
```

```
Number of LSA 3. Checksum Sum 0x017f02
```

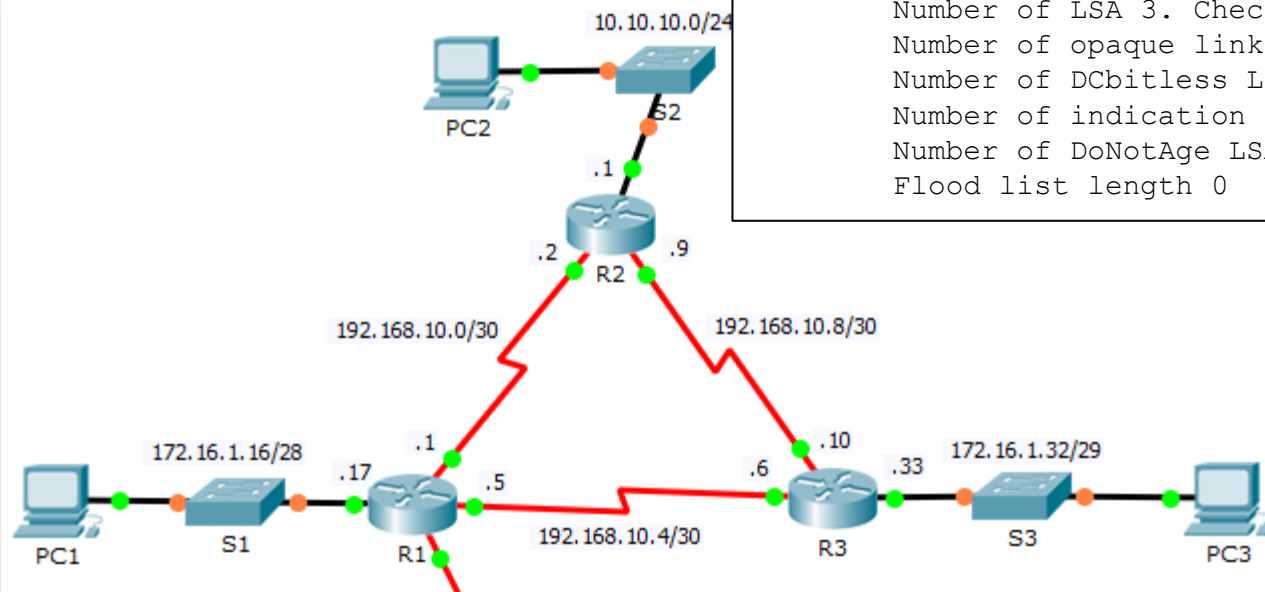
```
Number of opaque link LSA 0. Checksum Sum 0x000000
```

```
Number of DCbitless LSA 0
```

```
Number of indication LSA 0
```

```
Number of DoNotAge LSA 0
```

```
Flood list length 0
```

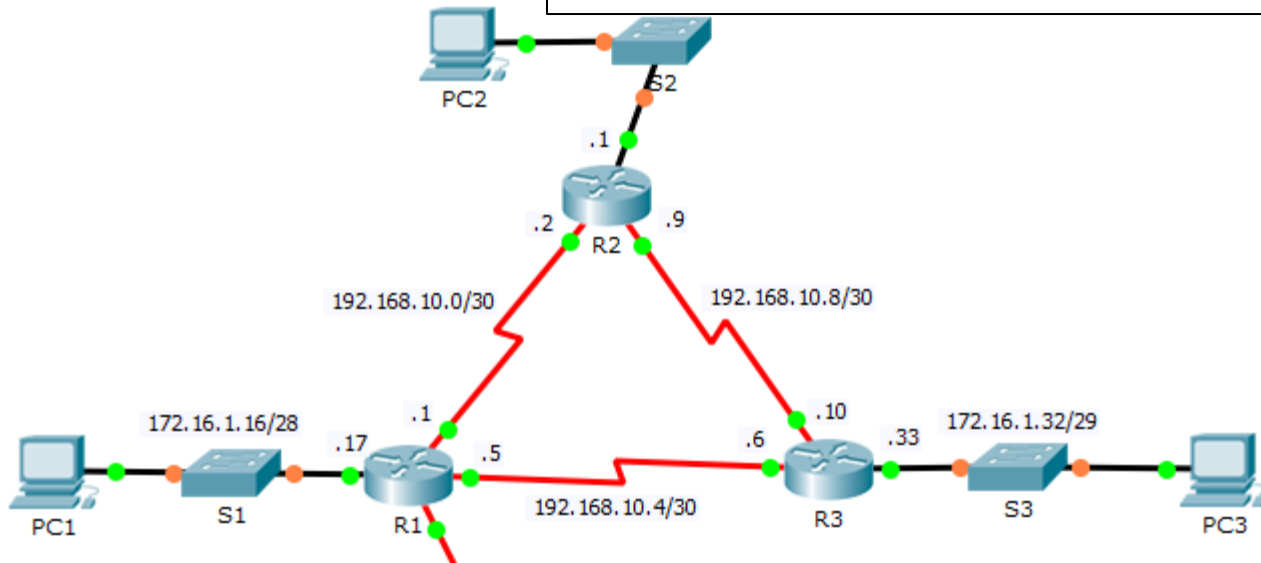


Verifying OSPF configuration

```
R1#show ip ospf interface se0/0/0
```

```
Serial0/0/0 is up, line protocol is up  
Internet address is 192.168.10.1/30, Area 0  
Process ID 1, Router ID 1.1.1.1, Network Type POINT-TO-POINT, Cost: 64  
Transmit Delay is 1 sec, State POINT-TO-POINT, Priority 0  
No designated router on this network  
No backup designated router on this network  
Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5  
Hello due in 00:00:05  
Index 2/2, flood queue length 0  
Next 0x0(0)/0x0(0)  
Last flood scan length is 1, maximum is 1  
Last flood scan time is 0 msec, maximum is 0 msec  
Neighbor Count is 1 , Adjacent neighbor count is 1  
Adjacent with neighbor 2.2.2.2  
Suppress hello for 0 neighbor(s)
```

```
R1#
```



Verifying OSPF configuration

```
R1#show ip ospf database
```

```
OSPF Router with ID (1.1.1.1) (Process ID 1)
```

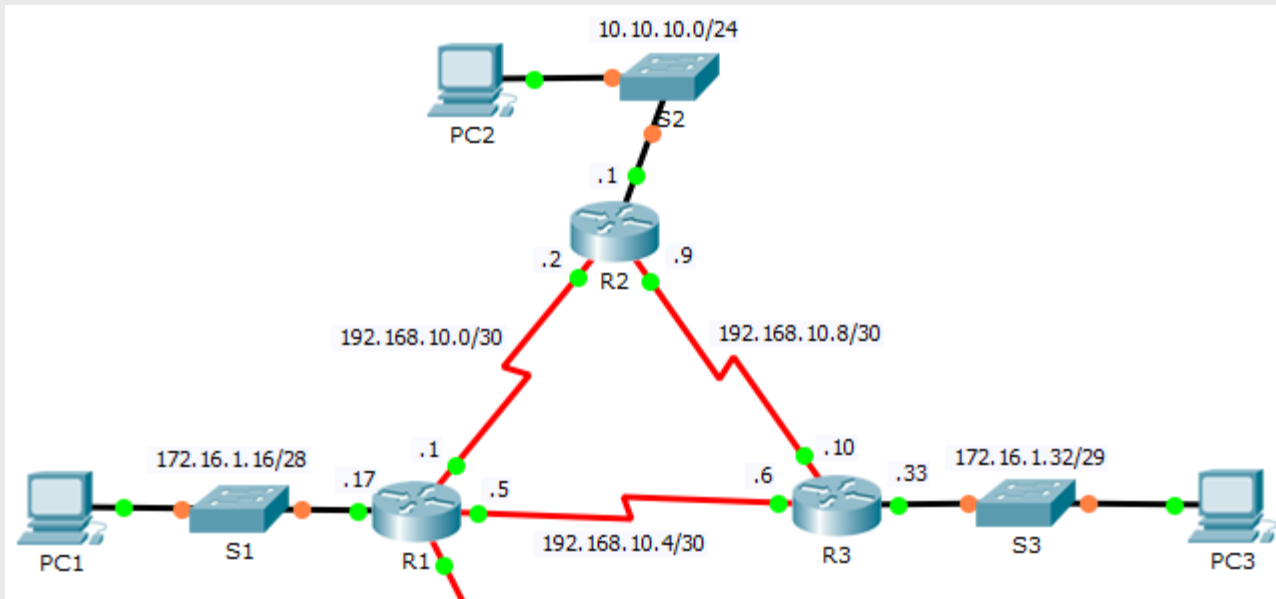
Router Link States (Area 0)

Link ID	ADV Router	Age	Seq#	Checksum	Link count
1.1.1.1	1.1.1.1	551	0x80000007	0x003367	5
2.2.2.2	2.2.2.2	551	0x80000006	0x00131e	5
3.3.3.3	3.3.3.3	551	0x80000006	0x0091d9	5

Type-5 AS External Link States

Link ID	ADV Router	Age	Seq#	Checksum	Tag
0.0.0.0	1.1.1.1	561	0x80000001	0x00fecf	1

```
R1#
```



Verifying OSPF configuration

```
R1#show ip ospf database router
```

```
OSPF Router with ID (1.1.1.1) (Process ID 1)
```

```
Router Link States (Area 0)
```

```
LS age: 116  
Options: (No TOS-capability, DC)  
LS Type: Router Links  
Link State ID: 1.1.1.1  
Advertising Router: 1.1.1.1  
LS Seq Number: 80000007  
Checksum: 0x3367  
Length: 84
```

AS Boundary Router

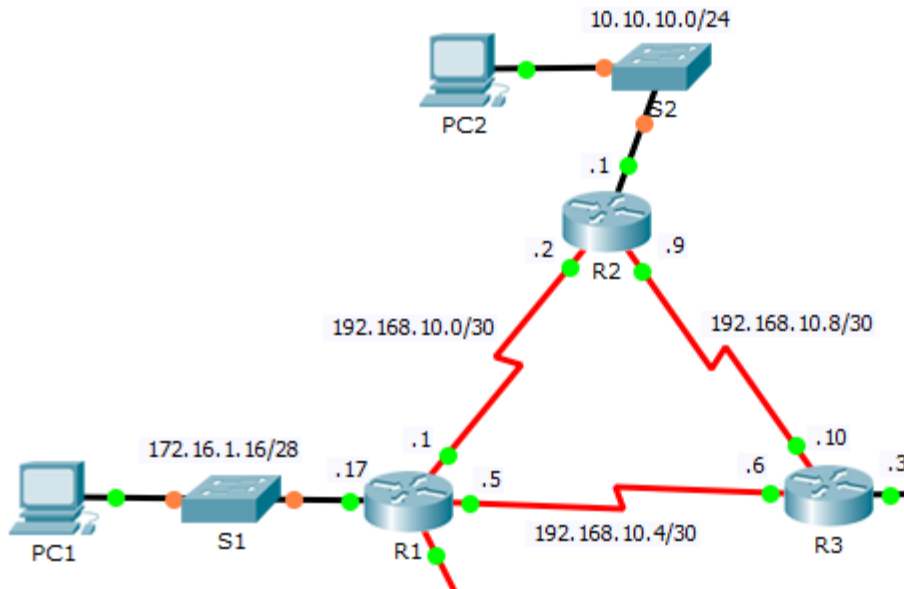
```
Number of Links: 5
```

```
Link connected to: a Stub Network  
(Link ID) Network/subnet number: 172.16.1.16  
(Link Data) Network Mask: 255.255.255.240  
Number of TOS metrics: 0  
TOS 0 Metrics: 1
```

```
Link connected to: another Router (point-to-point)  
(Link ID) Neighboring Router ID: 2.2.2.2  
(Link Data) Router Interface address: 192.168.10.1  
Number of TOS metrics: 0  
TOS 0 Metrics: 50
```

...

```
LS age: 116  
Options: (No TOS-capability, DC)  
LS Type: Router Links  
Link State ID: 3.3.3.3  
Advertising Router: 3.3.3.3  
LS Seq Number: 80000006  
Checksum: 0x91d9  
--More--
```



OSPF costs

■ Administrative Distance

Route Source	Administrative Distance
Connected	0
Static	1
EIGRP summary route	5
External BGP	20
Internal EIGRP	90
IGRP	100
OSPF	110
IS-IS	115
RIP	120
External EIGRP	170
Internal BGP	200

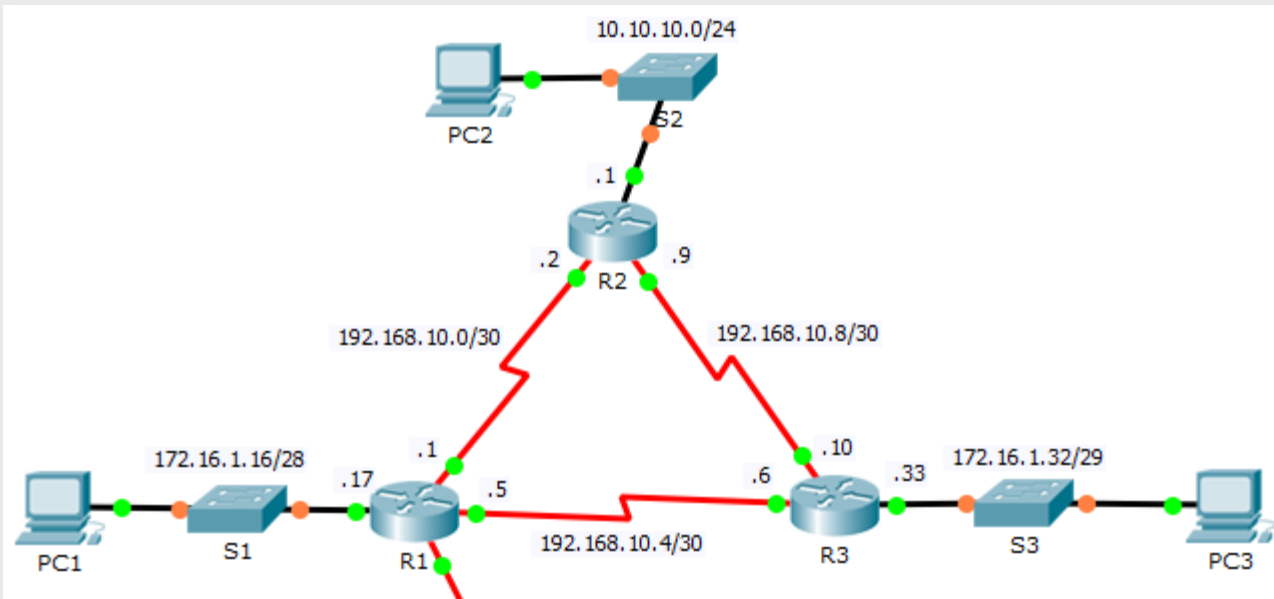
OSPF costs – IOS default

- Cisco IOS uses the bandwidth of the interface to calculate the OSPF cost:
 - **cost = *reference-bandwidth/interface-bandwidth***
- *reference-bandwidth* value is **100 Mbps** by default
 - can be changed with the OSPF command
 - `auto-cost reference-bandwidth reference-bandwidth`
- *bandwidth* value is determined by the interface configuration

Interface Type	100/bw in Mbps	Cost
10 Gigabit Ethernet	100/10,000	1
Gigabit Ethernet	100/1,000	1
Fast Ethernet	100/100	1
Ethernet	100/10	10
E1	100/2.048	48
T1	100/1.544	64
128 kbps	100/0.128	781
64 kbps	100/0.064	1562

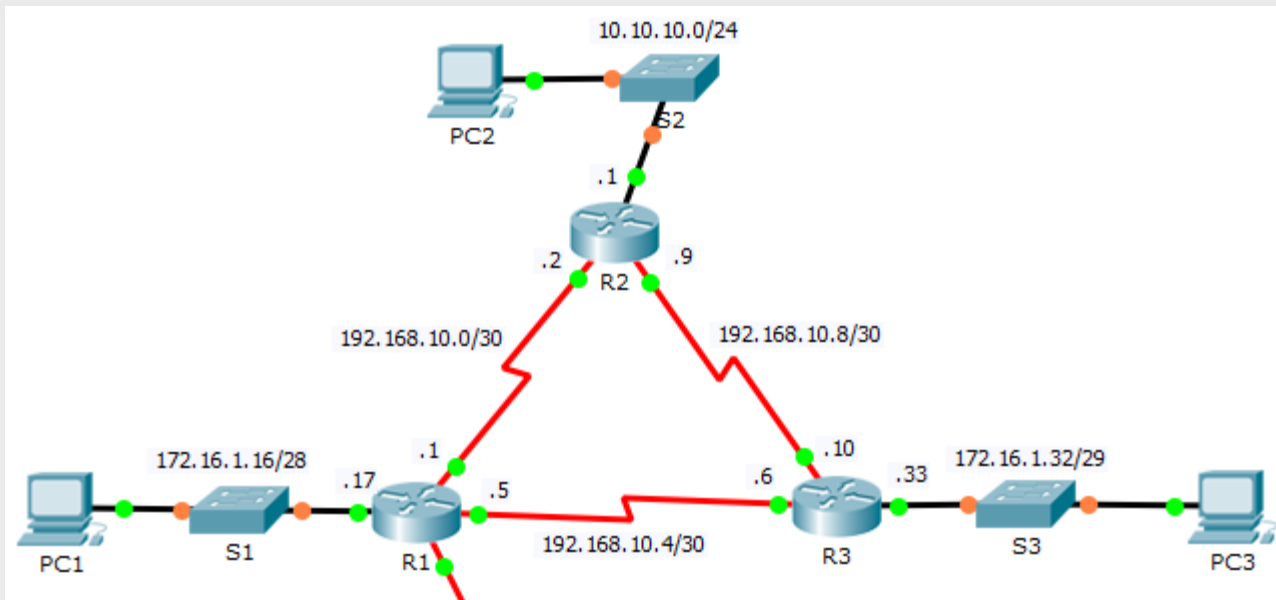
OSPF costs – cumulative cost

```
R1#show ip route
(**output omitted**)
O      10.10.10.0 [110/65] via 192.168.10.2, 01:07:18, Serial0/0/0
(**output omitted**)
R1#show ip route 10.10.10.0
Routing entry for 10.10.10.0/24
Known via "ospf 1", distance 110, metric 65, type intra area
  Last update from 192.168.10.2 on Serial0/0/0, 01:07:20 ago
  Routing Descriptor Blocks:
    * 192.168.10.2, from 2.2.2.2, 01:07:20 ago, via Serial0/0/0
      Route metric is 65, traffic share count is 1
R1#
```



Adjusting OSPF costs

- The actual speed of a link may be different than the default *bandwidth* value
 - Most serial links default to 1.544 Mbps
- The *bandwidth* value should be always configured to reflect the actual speed of the link
 - **Important:** this *bandwidth* value does not actually **affect** the speed of the link



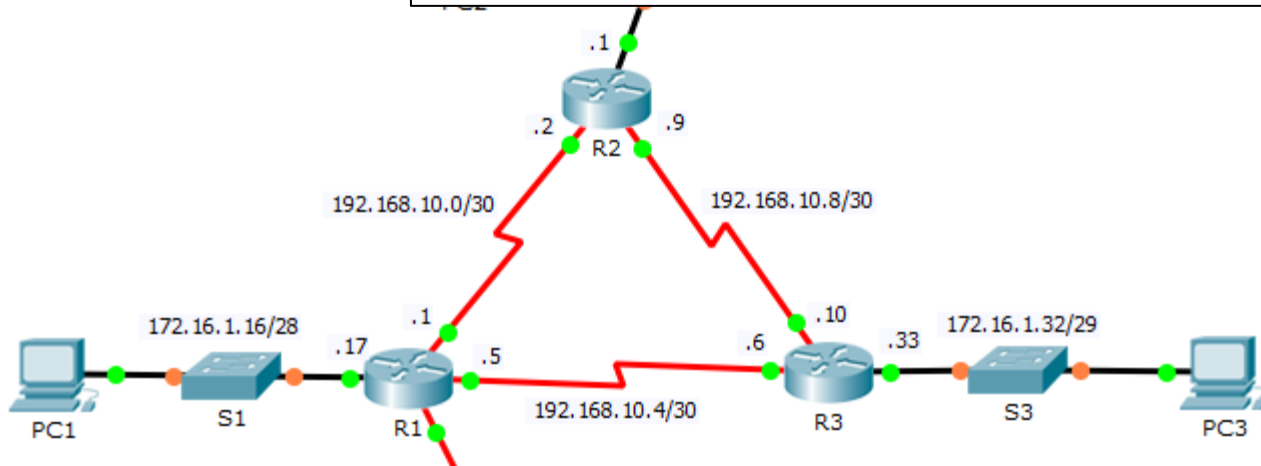
Adjusting OSPF costs

- The actual speed of a link may be different than the default *bandwidth* value
 - Most serial links
- The *bandwidth* value is not the actual speed of the link
 - Important to adjust the speed of the link

```
R1#show interface se0/0/0
(**output omitted**)
    MTU 1500 bytes, BW 1544 Kbit, DLY 20000 usec,
(**output omitted**)

R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#int se0/0/0
R1(config-if)#bandwidth 2000 ←
R1(config-if)#^Z
R1#show interface se0/0/0
(**output omitted**)
    MTU 1500 bytes, BW 2000 Kbit, DLY 20000 usec,
(**output omitted**)

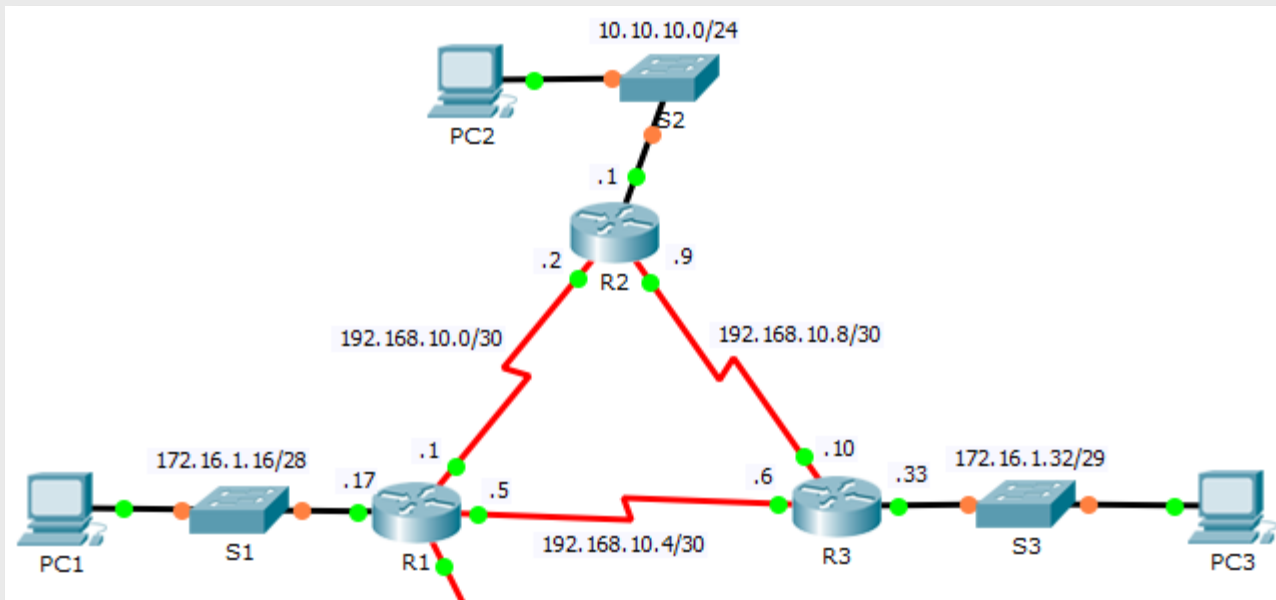
R1#show ip ospf interface se0/0/0
(**output omitted**)
    Process ID 1, Router ID 1.1.1.1, Network Type POINT-TO-POINT, Cost: 50
```



Adjusting OSPF costs

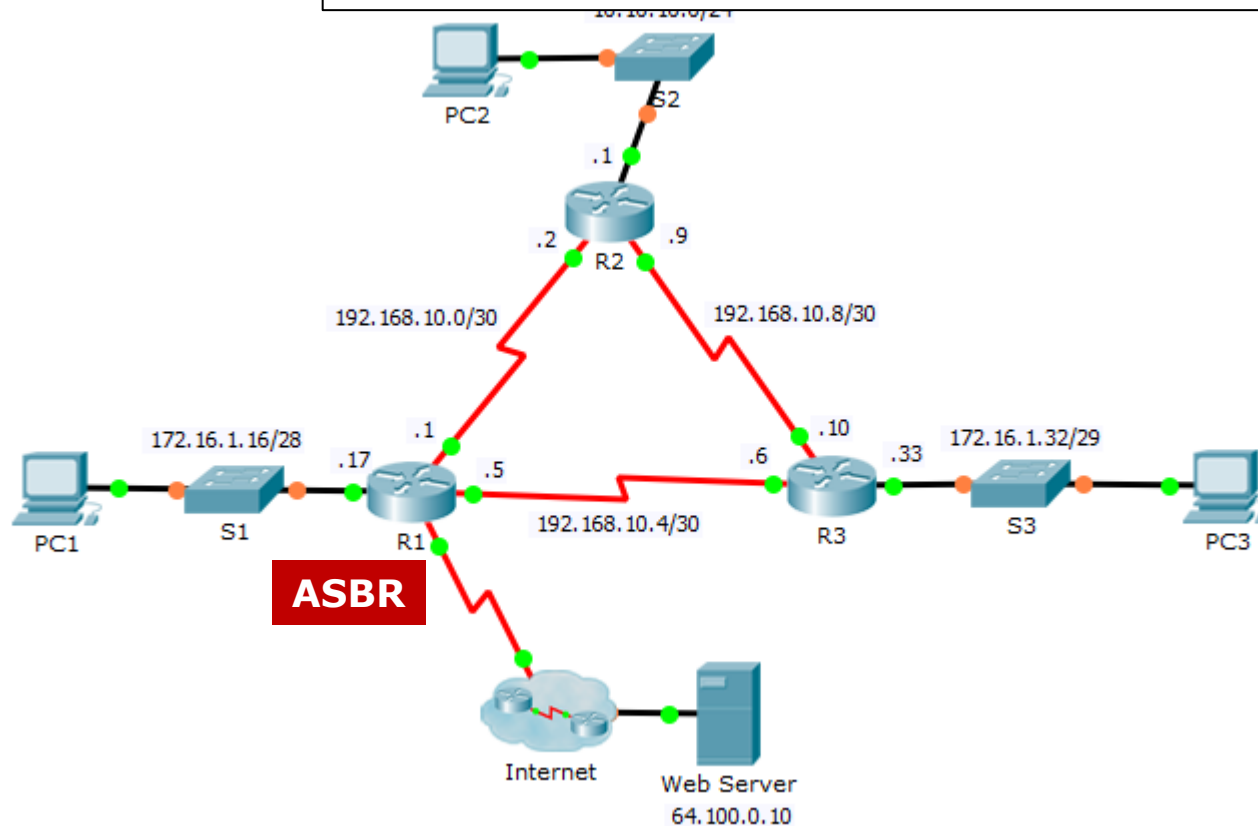
- Alternatively, costs can be administratively configured

```
R2(config)#int se0/0/0
R2(config-if)#ip ospf cost 50
R2(config-if)#^Z
R2#show interface se0/0/0
(**output omitted**)
      MTU 1500 bytes, BW 1544 Kbit, DLY 20000 usec,
(**output omitted**)
R2#show ip ospf interface se0/0/0
(**output omitted**)
      Process ID 1, Router ID 2.2.2.2, Network Type POINT-TO-POINT, Cost: 50
```



OSPF – redistributing a default route

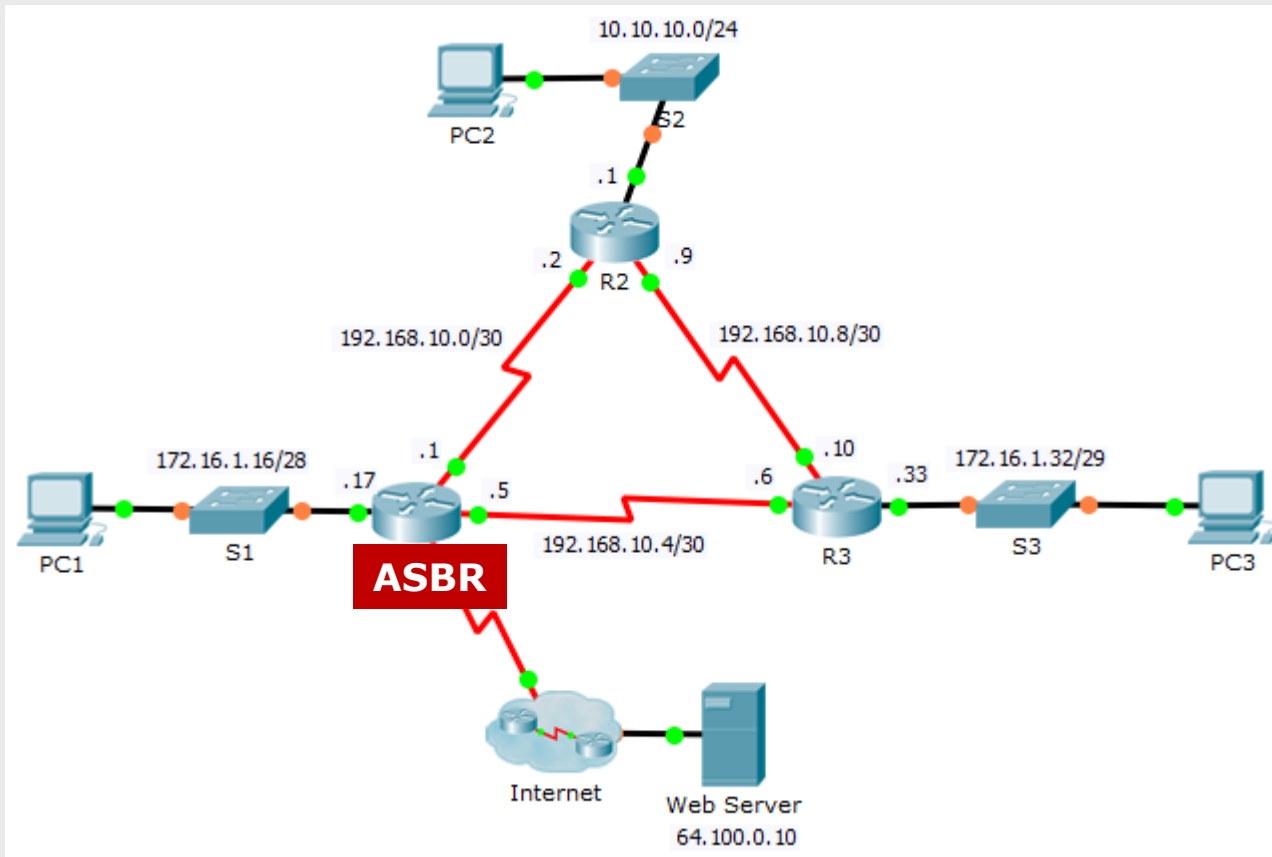
```
R1(config)#ip route 0.0.0.0 0.0.0.0 se0/1/0
R1(config)#router ospf 1
R1(config-router)#default-information originate
R1(config-router)#^Z
R1#
```



OSPF – redistributing a default route

```
R2#show ip route
(**output omitted**)
Gateway of last resort is 192.168.10.1 to network 0.0.0.0

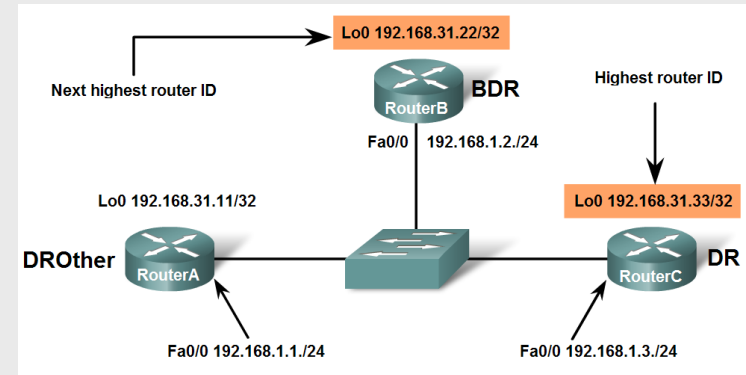
(**output omitted**)
O*E2 0.0.0.0/0 [110/1] via 192.168.10.1, 00:00:17, Serial10/0/0
R2#
```



OSPF – multiaccess networks

■ Designated Router (DR) and Backup Designated Router (BDR) election

- **DR:** Router with the **highest** OSPF interface **priority**
- **BDR:** Router with the **second highest** OSPF interface **priority**
 - Router(config-if)#**ip ospf priority** {0 - 255}
- If OSPF interface priorities are equal, the **highest router ID** is used to break the tie



```
RouterA#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
192.168.31.33	1	FULL/DR	00:00:39	192.168.1.3	FastEthernet0/0
192.168.31.22	1	FULL/BDR	00:00:36	192.168.1.2	FastEthernet0/0

```
RouterB#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
192.168.31.33	1	FULL/DR	00:00:34	192.168.1.3	FastEthernet0/0
192.168.31.11	1	FULL/DROTHER	00:00:38	192.168.1.1	FastEthernet0/0

```
RouterC#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
192.168.31.22	1	FULL/BDR	00:00:35	192.168.1.2	FastEthernet0
192.168.31.11	1	FULL/DROTHER	00:00:32	192.168.1.1	FastEthernet0

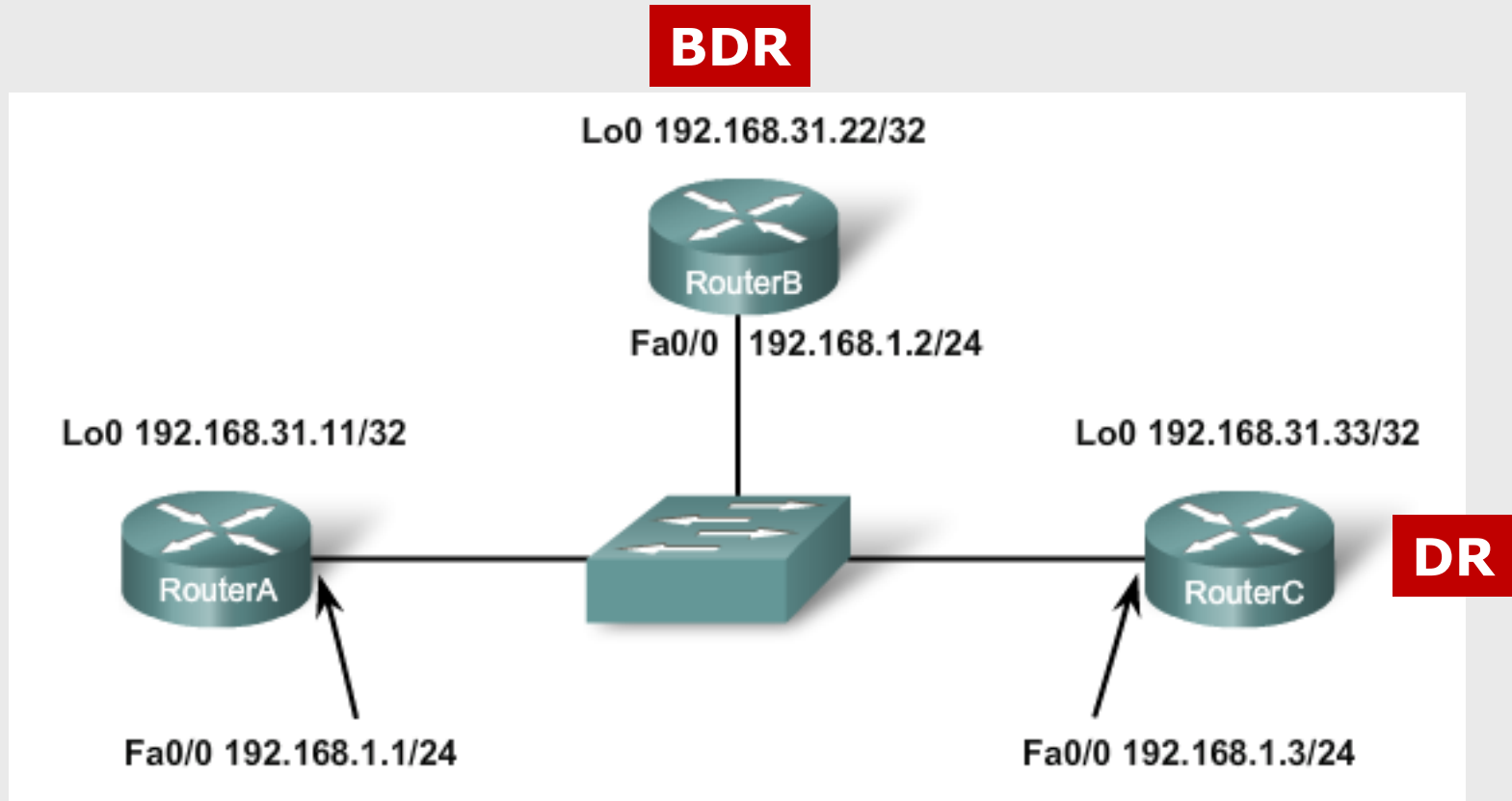
Priority is equal at the default value of 1.

```
RouterA#show ip ospf interface fastethernet 0/0
```

```
FastEthernet0/0 is up, line protocol is up
Internet Address 192.168.1.1/24, Area 0
Process ID 1, Router ID 192.168.31.11, Network Type BROADCAST, Cost: 1
Transmit Delay is 1 sec, State DROTHER, Priority 1
Designated Router (ID) 192.168.31.33, Interface address 192.168.1.3
Backup Designated router (ID) 192.168.31.22, Interface address 192.168.1.2
Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
  oob-resync timeout 40
  Hello due in 00:00:06
Supports Link-local Signaling (LLS)
Index 1/1, flood queue length 0
Next 0x0(0)/0x0(0)
Last flood scan length is 0, maximum is 1
Last flood scan time is 0 msec, maximum is 0 msec
Neighbor Count is 2, Adjacent neighbor count is 2
  Adjacent with neighbor 192.168.31.22 (Backup Designated Router)
  Adjacent with neighbor 192.168.31.33 (Designated Router)
Suppress hello for 0 neighbor(s)
```

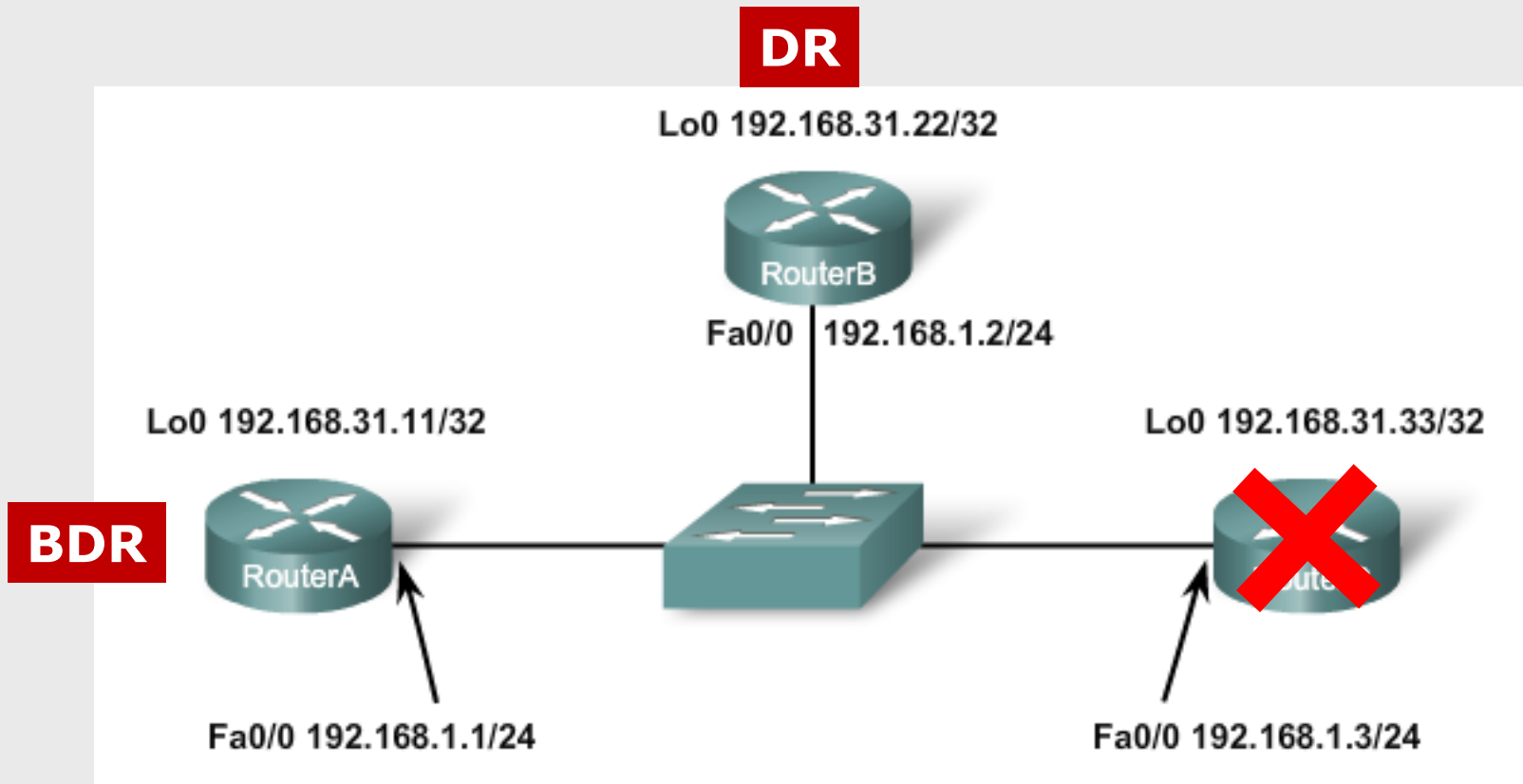
OSPF – multiaccess networks

- Election takes place as soon as the first router is active
 - It is possible that a router with a lower Router ID becomes the DR
- When the DR is elected, it remains until it fails



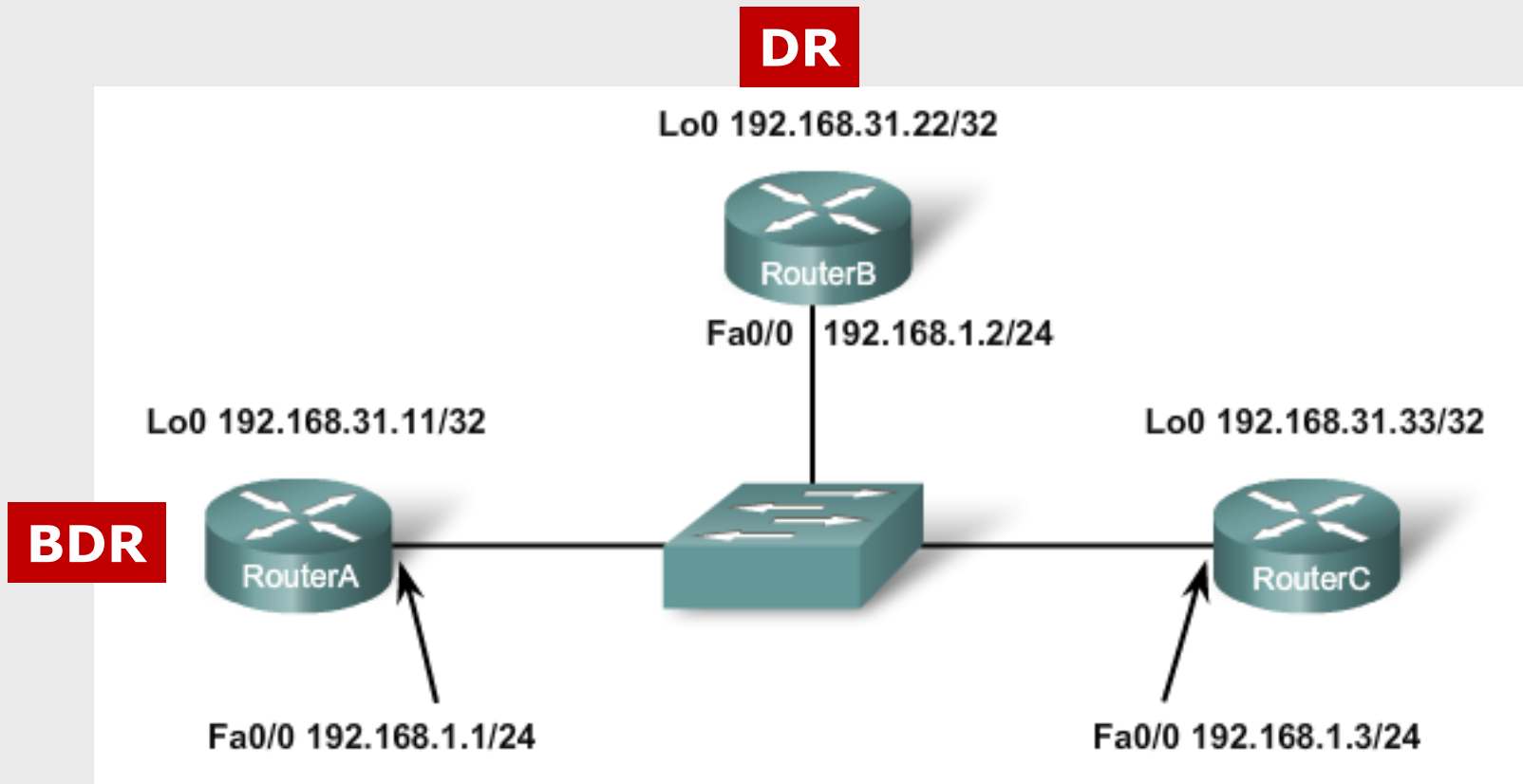
OSPF – multiaccess networks

- Election takes place as soon as the first router is active
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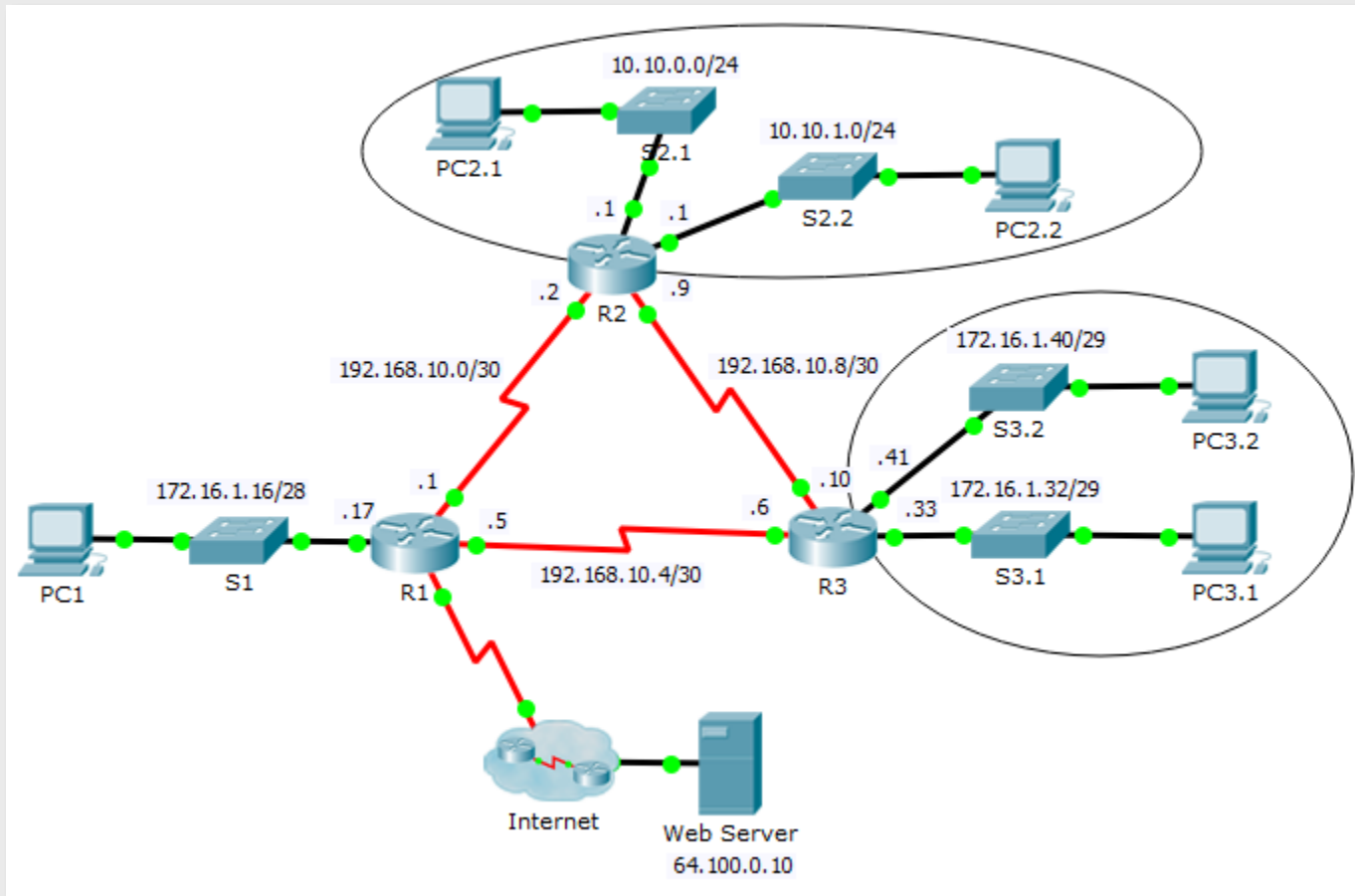
OSPF – multiaccess networks

- Election takes place as soon as the first router is active
 - It is possible that a router with a lower Router ID becomes the DR
- When the DR is elected, it remains until it fails



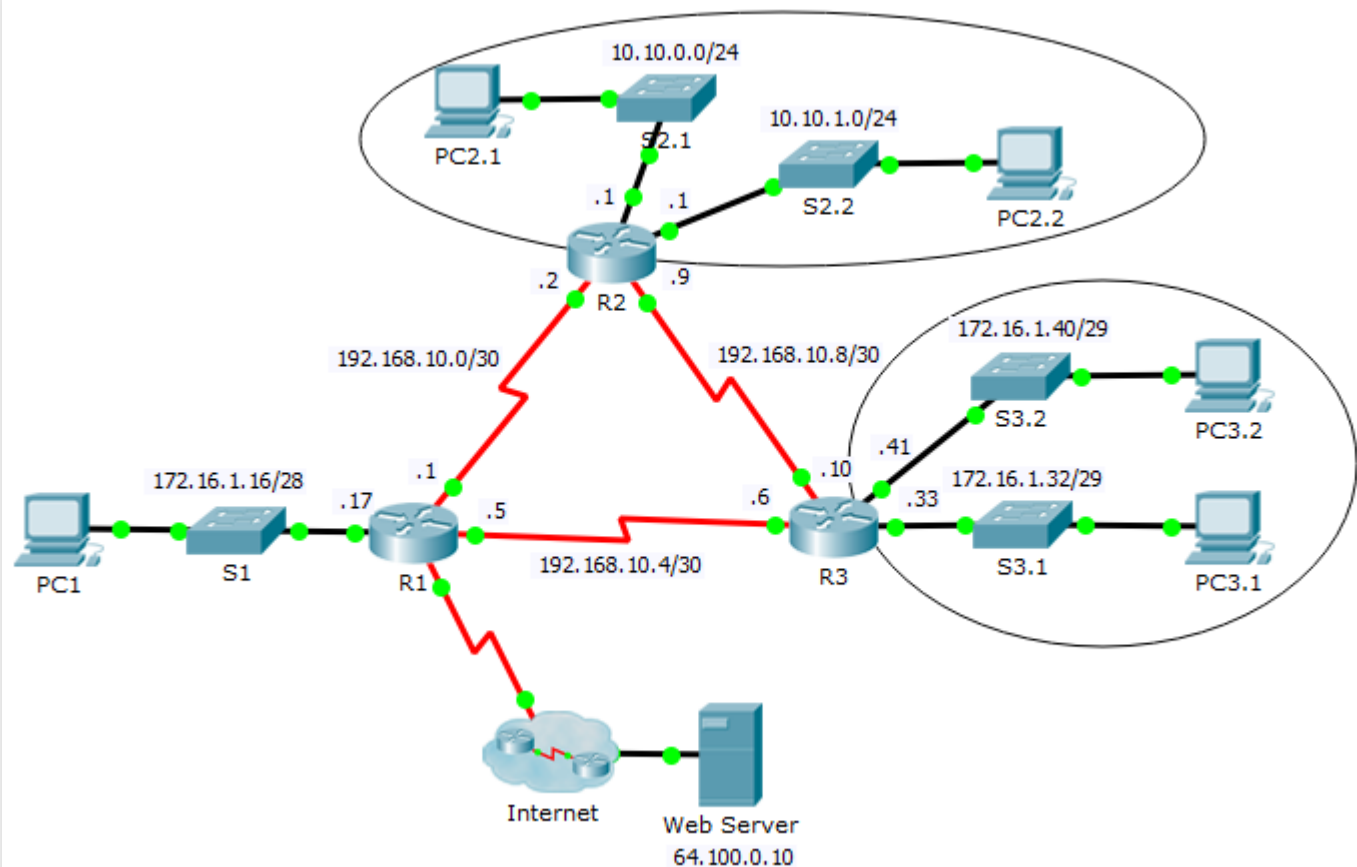
IOS OSPF configuration – multi area

■ Reference topology



IOS OSPF configuration

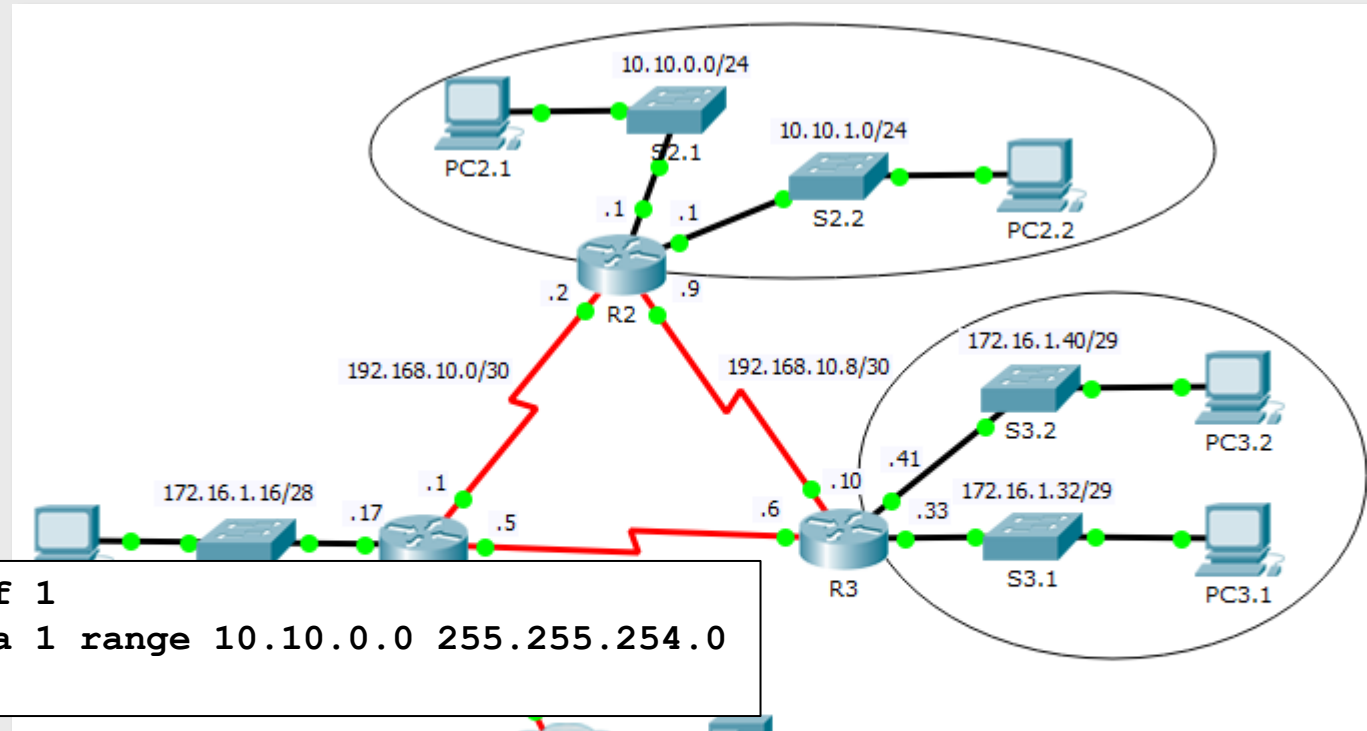
```
R2 (config)#router ospf 1  
R2 (config-router)#network 192.168.10.2 0.0.0.0 area 0  
R2 (config-router)#network 192.168.10.9 0.0.0.0 area 0  
R2 (config-router)#network 10.10.0.1 0.0.0.0 area 1  
R2 (config-router)#network 10.10.1.1 0.0.0.0 area 1  
R2 (config-router)#^Z
```



OSPF inter-area route summarization

- OSPF does not perform auto-summarization
- Inter-area summarization must be manually configured on ABRs

```
Router(config-router)#area area-id range address mask
```

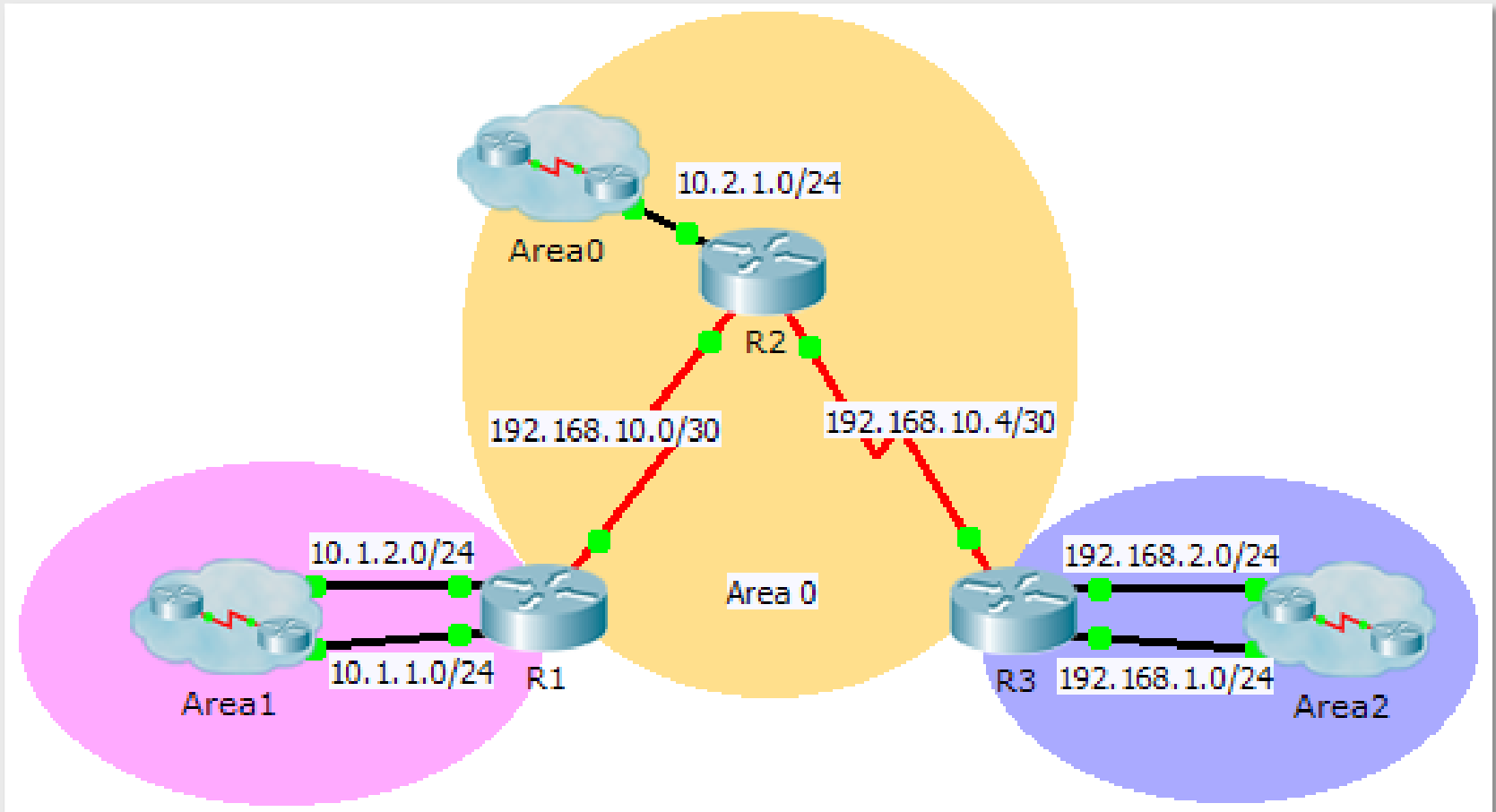


```
R2(config)#router ospf 1  
R2(config-router)#area 1 range 10.10.0.0 255.255.254.0  
R2(config-router)#^Z
```

```
R3(config)#router ospf 1  
R3(config-router)#area 2 range 172.16.1.32 255.255.255.240  
R3(config-router)#^Z
```

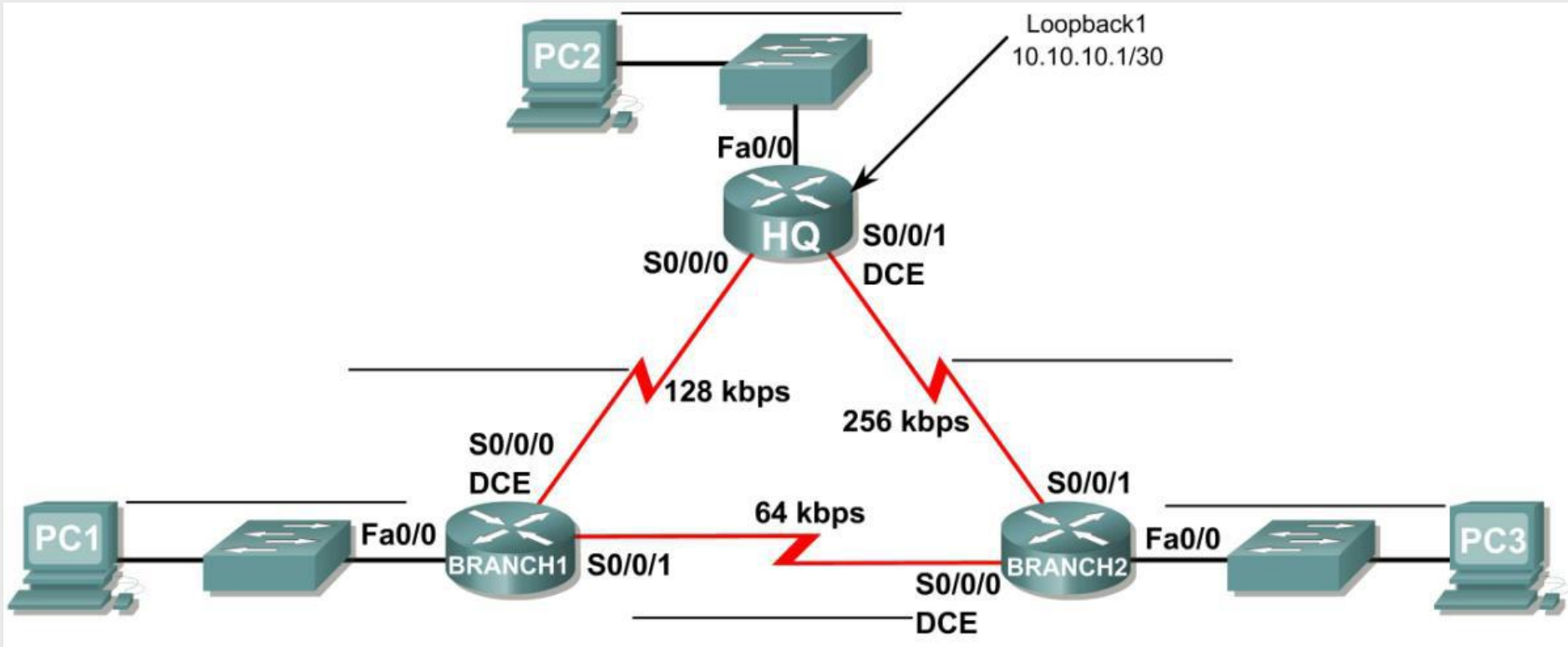
Lab activity

■ OSPF multi-area



Lab activity

■ OSPF – single area



Lab activity

- OSPF – multi area

